

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, TypeScript, C++, JavaScript, SQL

Tools/Frameworks: React, Flask, FastAPI, Docker, Kubernetes, TensorFlow, PyTorch, PostgreSQL, AWS SageMaker

Competence: Full-stack Development, Machine Learning, MLOps, LLM Integration, AI Applications, Data Engineering, CI/CD, Containerization

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, enhancing model performance and training efficiency.
- Developed physics-informed hybrid loss functions and explored novel architectures to improve model accuracy and stability.
- Collaborated with PhD researchers to co-author a publication presented at the FLOMEKO conference.
- Built interactive dashboards for data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased processing speed by 5x using AI algorithms, deployed with Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure.
- Created POCs using YOLO, few-shot learning, and CNNs for client demonstrations.
- Conducted educational sessions on machine learning and big data at universities, mentoring students in AI applications.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application for AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and LLM tools to enhance user engagement.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented web scrapers and ETL pipelines for structured and unstructured data.
- Automated data ingestion and cleaning workflows to optimize client analytics processes.

Projects

CAFA 6: Protein Function Prediction 2022

Deep Learning Competition

- Engineered a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Achieved a 4% increase in model accuracy across MF, CC, and BP namespaces through advanced feature engineering techniques.

Razanlytics: Real-time Bitcoin Price Forecasting 2022

Thesis Project

- Developed a real-time Bitcoin price forecasting system using machine learning models and time-series analysis.
- Implemented a scalable architecture for data collection, preprocessing, and prediction.

Saaz: AI Self-Playing Chitralli Sitar 2023

Signal Processing, Music Generation

- Built the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI applications in music.

Hugging Face Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität Present

Master of Science in Artificial Intelligence *Erlangen, DE*

- Specializing in AI applications, full-stack development, and LLM integration.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor.
- Focused on machine learning, data engineering, and software development.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python